

- 7 (a) deviation shown correctly B1 [1]
- (b) smaller deviation (not zero deviation) M1
acceptable path wrt position of N A1 [2]
- (c) the nucleus is (very) small M1
in comparison to the atom A1 [2]
(special case: 'atom is mostly empty space' scores 1 mark)
- (d) deviation depends on charge on the nucleus / N / electrostatic repulsion B1
same charge so no change in deviation B1 [2]

[Total: 7]